

# Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

**Benchmark:** MENTALCLOUDS | **Context:** Global South Mental Health Practitioners — India-Anchored OMHC / Peer-Support Deployment

**Input Ontology definition:** Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

The benchmark's three-component taxonomy (symptom and history exploration, patient discovery, reflective utterances) **[Q15]** is structurally constrained relative to the clinical documentation needs of the target deployment. The authors themselves acknowledge that 'only three aspects (counseling component) of the conversation' were explored **[Q69]**, that models cannot identify psychotherapy types or therapy techniques **[Q63]**, and that 'important negative histories... such as the history of suicide risk or substance use were also not recorded' **[Q64]**. For the India-anchored deployment, MHCA 2017 Section 115 establishes a duty-of-care around suicide risk **[WEB-10, WEB-11]**, and India's National Suicide Prevention Strategy 2022 **[WEB-30]** makes suicide-risk capture a national policy priority — yet the benchmark ontology has no category for it. The elicitation explicitly identifies risk flags, family-related suggestions, and spiritual/religious coping as required content (A2), all of which fall outside the three-component frame. The authors also flag that the three components 'frequently overlap, posing clinical and legal problems' **[Q62]**, indicating internal incoherence of the taxonomy itself. Given IO is HIGH priority and multiple critical regional categories are absent, the dimension is significantly misaligned.

| ID            | Evidence                                                                                                                                                                                                                                |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q15</b>    | "Within each conversation, three pivotal counseling components (aspects) emerge – symptom and history exploration, patient discovery, and reflective utterances." (p.6)                                                                 |
| <b>Q62</b>    | "However, the models did not do as well with the structure separation of the information. The sections of "symptoms and history", "patient discovery", and "reflection" frequently overlap, posing clinical and legal problems." (p.15) |
| <b>Q63</b>    | "The models are also unable to identify psychotherapy types (e.g., cognitive behavior therapy) and therapy techniques, which form an integral part of counseling notes." (p.15)                                                         |
| <b>Q64</b>    | "Important negative histories gathered during the session, such as the history of suicide risk or substance use were also not recorded, and in at least one instance, the presence of suicide risk was not identified." (p.15)          |
| <b>Q69</b>    | "Finally, this work explored only three aspects (counseling component) of the conversation." (p.16)                                                                                                                                     |
| <b>WEB-10</b> | MHCA 2017 Section 115 imposes duty-of-care on suicide risk ( <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC6482674/">https://pmc.ncbi.nlm.nih.gov/articles/PMC6482674/</a> )                                                        |
| <b>WEB-11</b> | <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC5914247/">https://pmc.ncbi.nlm.nih.gov/articles/PMC5914247/</a>                                                                                                                       |
| <b>WEB-30</b> | India National Suicide Prevention Strategy 2022 ( <a href="https://www.pib.gov.in/PressReleasePage.aspx?PRID=2188003">https://www.pib.gov.in/PressReleasePage.aspx?PRID=2188003</a> )                                                   |

## Strengths

- Aspect-based decomposition into distinct counseling components is a meaningful conceptual advance over generic dialogue summarization **[Q1, Q19, Q71]**, and the three components captured do reflect core CBT-aligned clinical reasoning steps that the primary CBT-trained practitioner cohort (A3) would recognize.
- Symptom-and-history and reflective utterances align reasonably with the CBT-anchored structure of the primary practitioner cohort's documentation needs **[Q15, A3]**.

| ID         | Evidence                                                                                                                                                                                                                                     |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q1</b>  | "This study evaluates the effectiveness of state-of-the-art Large Language Models (LLMs) in selectively summarizing various components of therapy sessions through aspect-based summarization, aiming to benchmark their performance." (p.1) |
| <b>Q15</b> | "Within each conversation, three pivotal counseling components (aspects) emerge – symptom and history exploration, patient discovery, and reflective utterances." (p.6)                                                                      |

|            |                                                                                                                                             |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q19</b> | "This section reports the aspect-based (psychotherapy element-based) summarization results based on the automatic evaluation scores." (p.9) |
| <b>Q71</b> | "Our study benchmarked the efficacy and role of large language models towards component-guided counseling summarization tasks." (p.16)      |

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                            |
|-------------|----------------------------------------------------------------|
| <b>IO-1</b> | Identify test case categories required for regional deployment |
| <b>IO-2</b> | Check if taxonomy omits regionally relevant categories         |
| <b>IO-3</b> | Check if taxonomy includes irrelevant categories               |
| <b>IO-4</b> | Document category gaps harming content validity                |

### Checklist responses:

**IO-1:** Required categories for the deployment include the three benchmark components plus risk-and-safety flagging (suicidal ideation, self-harm, substance use), family-directed interventions, spiritual/religious coping, and therapy-technique identification — established by elicitation (A2) and reinforced by Indian regulatory context (MHCA 2017 Section 115 [WEB-10]; National Suicide Prevention Strategy [WEB-30]).

**IO-2:** Yes — the taxonomy omits suicide/self-harm risk and substance-use history [Q64], psychotherapy-type and therapy-technique identification [Q63], family-directed interventions (no quote; structural absence confirmed; MHCA 2017 itself critiqued for inadequate family-caregiver framing [WEB-9]), and spiritual/religious coping (American-sourced sessions cannot represent Indian faith-healer integration patterns [WEB-15]).

**IO-3:** No clearly irrelevant categories were identified; the three components are clinically meaningful, though their frequent overlap [Q62] introduces construct-irrelevant variance.

**IO-4:** Documented gaps that harm content validity: (a) absence of risk-flag category despite legal and policy duty in India; (b) absence of family-intervention category for South Asian relational documentation needs; (c) absence of spiritual-coping category given documented faith-healer integration [WEB-15]; (d) absence of therapy-technique identification [Q63]; (e) overlap between the three existing components [Q62].

| ID            | Evidence                                                                                                                                                                                                                                |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q62</b>    | "However, the models did not do as well with the structure separation of the information. The sections of "symptoms and history", "patient discovery", and "reflection" frequently overlap, posing clinical and legal problems." (p.15) |
| <b>Q63</b>    | "The models are also unable to identify psychotherapy types (e.g., cognitive behavior therapy) and therapy techniques, which form an integral part of counseling notes." (p.15)                                                         |
| <b>Q64</b>    | "Important negative histories gathered during the session, such as the history of suicide risk or substance use were also not recorded, and in at least one instance, the presence of suicide risk was not identified." (p.15)          |
| <b>WEB-9</b>  | MHCA 2017 critiqued for not adequately addressing family caregiver roles ( <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC8277537/">https://pmc.ncbi.nlm.nih.gov/articles/PMC8277537/</a> )                                          |
| <b>WEB-10</b> | MHCA 2017 Section 115 imposes duty-of-care on suicide risk ( <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC6482674/">https://pmc.ncbi.nlm.nih.gov/articles/PMC6482674/</a> )                                                        |
| <b>WEB-15</b> | Faith-healer integration documented in Indian mental health practice ( <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC7616029/">https://pmc.ncbi.nlm.nih.gov/articles/PMC7616029/</a> )                                              |
| <b>WEB-30</b> | India National Suicide Prevention Strategy 2022 ( <a href="https://www.pib.gov.in/PressReleasePage.aspx?PRID=2188003">https://www.pib.gov.in/PressReleasePage.aspx?PRID=2188003</a> )                                                   |

## Information Gaps

- Whether reference summaries informally mention risk or family content even though no scoring component captures them — would require dataset-level inspection.

## Requires Expert Verification

- Clinical confirmation from Indian OMHC practitioners that risk, family, and spiritual categories are universally required (vs. context-dependent).
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## Remediation

**Gap 1: No taxonomy categories for suicide/self-harm risk, substance use, family-directed interventions, spiritual/religious coping, or therapy-technique identification.**

**Recommendation:** Extend the aspect taxonomy with at least three additional components — risk and safety, family/relational interventions, and spiritual/religious coping — and add therapy-technique identification as a sub-task. Tie the risk component explicitly to MHCA 2017 Section 115 documentation obligations.